

Suite Hard • Simulation Module 13

22 questions • 35 minutes • official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

Read the prompt carefully. Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 6

Advanced Math: 6

PSDA: 5

Geometry: 5

Do not open the next module until this one is timed and scored.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

Jeremy deposited x dollars in his investment account on January 1, 2001. The amount of money in the account doubled each year until Jeremy had 480 dollars in his investment account on January 1, 2005. What is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The equation $\frac{x^2+6x-7}{x+7} = ax+d$ is true for all $x \neq -7$, where a and d are integers. What is the value of $a+d$?

Answer

- A. -6
- B. -1
- C. 0
- D. 1

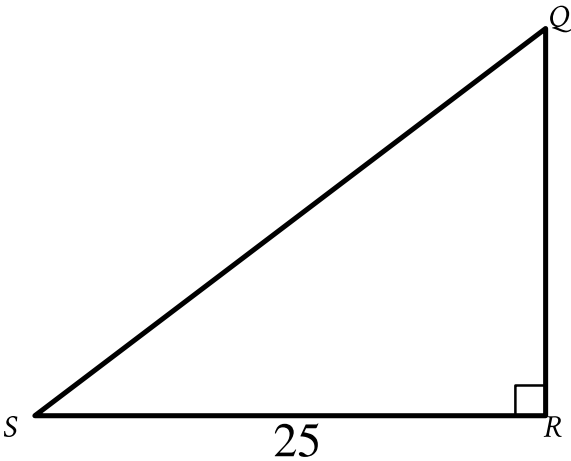
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $25 \cos Q$

B. $25 \sin Q$

C. $\frac{25}{\cos Q}$

D. $\frac{25}{\sin Q}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A sample of 40 fourth-grade students was selected at random from a certain school. The 40 students completed a survey about the morning announcements, and 32 thought the announcements were helpful. Which of the following is the largest population to which the results of the survey can be applied?

Answer

- A. The 40 students who were surveyed
- B. All fourth-grade students at the school
- C. All students at the school
- D. All fourth-grade students in the county in which the school is located

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x - 14)(x + 19)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Answer

- A. -266
- B. -19
- C. $-\frac{33}{2}$
- D. $-\frac{5}{2}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

$$0.10x + 0.20y = 0.17(x + y)$$

The equation gives a volume x , in gallons, of a **10%** solution that could be mixed with a volume y , in gallons, of a **20%** solution to produce a **17%** solution. According to this equation, what volume, in gallons, of the **20%** solution could be mixed with **90.0** gallons of the **10%** solution to produce a **17%** solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$x^2 + 20x + y^2 + 16y = -20$

The equation above defines a circle in the xy-plane. What are the coordinates of the center of the circle?

- Answer**
- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

The density of a certain type of wood is **353** kilograms per cubic meter. A sample of this type of wood is in the shape of a cube and has a mass of **345** kilograms. To the nearest hundredth of a meter, what is the length of one edge of this sample?

Answer

- A. 0.98
- B. 0.99
- C. 1.01
- D. 1.02

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$(7532 + 100y^2) + 10(10y^2 - 110)$

The expression above can be written in the form $ay^2 + b$, where a and b are constants. What is the value of $a + b$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$x + y = 20$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

Answer

- A. $3x + 3y = 60$
- B. $-x - y = 40$
- C. $3x + 3y = 14$
- D. $x - y = 20$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of $\overline{AB}$ is 2.9 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of **27** values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Answer

- A. The mean of data set A is greater than the mean of data set B.
- B. The mean of data set A is less than the mean of data set B.
- C. The mean of data set A is equal to the mean of data set B.
- D. There is not enough information to compare the means of the data sets.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$f(x) = x^3 - 9x$

$g(x) = x^2 - 2x - 3$

Which of the following expressions is equivalent to $\frac{f(x)}{g(x)}$, for $x > 3$?

- Answer
- A. $\frac{1}{x+1}$
- B. $\frac{x+3}{x+1}$
- C. $\frac{x(x-3)}{x+1}$
- D. $\frac{x(x+3)}{x+1}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

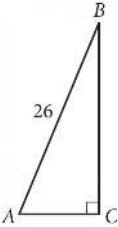
Question

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

- Answer**
- A. $5x$
 - B. $10x$
 - C. $2x + 20$
 - D. $10x + 20$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

Anita created a batch of green paint by mixing 2 ounces of blue paint with 3 ounces of yellow paint. She must mix a second batch using the same ratio of blue and yellow paint as the first batch. If she uses 5 ounces of blue paint for the second batch, how much yellow paint should Anita use?

Answer

- A. Exactly 5 ounces
- B. 3 ounces more than the amount of yellow paint used in the first batch
- C. 1.5 times the amount of yellow paint used in the first batch
- D. 1.5 times the amount of blue paint used in the second batch

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y , the table gives three values of x and their corresponding values of y , where t is a constant. Which equation represents this relationship?

- Answer
- A. $y = -2x + t + 23$

B. $y = 2x + t + 23$

C. $y = -\frac{23}{17}x + t + 46$

D. $y = \frac{23}{17}x + t + 46$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y + 7 = 7$$

$$20x + 100 - y = 0$$

The solution to the given system of equations is (x, y) . What is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

Module 13 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

21. _____

22. _____